Contrastive vs Negative-Free Self-Supervised Learning: A SimCLR and SimSiam Comparison

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***Abstract*—Self-supervised learning (SSL) has emerged as a powerful model for learning visual representations from unlabeled data and offers a practical alternative to supervised pretraining when labeled examples are scarce or difficult to obtain. This paper presents an implementation and evaluation of two prominent SSL methods; SimCLR, a contrastive approach, and SimSiam, a distillation-based approach without negative pairs on the *STL10 dataset* downscaled to a 32x32 resolution. We pretrain both methods on 100,000 unlabeled images and evaluate the learned representations via linear probing with varying label budgets (10 to 500 images per class), which we compare against a fully supervised baseline that was trained end-to-end on the labeled subset. Our results show that both SSL methods produce representations that match the supervised baseline under linear evaluation, achieving 66% test accuracy at 100 training epochs against 67% for the baseline, with SimSiam and SimCLR performing within one percentage point of each other. One thing to note is that, while the supervised baseline reaches 98% training accuracy by 500 epochs, which indicates severe overfitting on the small labeled set, the SSL-trained linear classifiers exhibit far better train-test agreement. These findings confirm that SSL can produce competitive representations on small-scale, low-resolution benchmarks, although the dramatic improvements reported for large-scale SSL are not fully realized in this compute constrained environment.**

***Index Terms*—Computer Vision, Self-Supervised Learning, Contrastive Learning, Representation Learning, Linear Evaluation, Data Augmentation**

I. Introduction

MODERN deep learning has achieved remarkable success on visual recognition tasks, but these successes typically depend on large labeled datasets. In order to produce these extensive datasets, it is very expensive and time consuming. For example, collecting 100,000 images for an autonomous driving application might require an hour of driving, but annotating those same images with pixel-level semantic segmentation labels can take months of human effort. This asymmetry between the cost of data collection and the cost of annotation has motivated a growing body of research into self-supervised learning (SSL), which seeks to learn useful representations directly from unlabeled data. A model pretrained using SSL can subsequently be fine-tuned on a small labeled dataset, offering a practical alternative to transfer learning from ImageNet-pretrained models.

Among SSL methods for images, contrastive learning has proven especially effective. Contrastive methods train a model to produce similar representations for two augmented views of the same image while producing dissimilar representations for view of different images. SimCLR is a widely adopted instantiation of this idea. It combines a convolutional encoder, a small MLP projection head, and the InfoNCE loss applied over pairs of augmented views within a batch. By forcing the encoder to capture features invariant to the chosen augmentations — cropping, color jitter, blurring, and so on — the method learns representations that transfer well to downstream classification.

More recently, alternative SSL architectures have emerged that avoid the need for explicit negative pairs. Distillation-based methods such as BYOL (Bootstrap Your Own Latent) and SimSiam (Simple Siamese) use an asymmetric architecture and a stop-gradient operation to prevent representation collapse, while redundancy-reduction methods such as Barlow Twins regularize the cross-correlation matrix of twin network outputs. SimSiam is particularly appealing from an implementation standpoint because it removes both the momentum encoder of BYOL and the negative sampling of SimCLR,

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relying solely on a predictor network and stop-gradient to achieve competitive performance.

In this paper, we implement and evaluate both SimCLR and SimSiam on the STL10 dataset. Our contributions are threefold. First, we implement the SimCLR framework with a compact CNN encoder and a two layer MLP projection head trained with the InfoNCE loss on 100,000 unlabeled STL10 images. Second, we evaluate the learned representations via logistic regression across label budgets ranging from 10 to 500 images per class and compare against a fully supervised baseline trained end-to-end on the same labeled subsets. Third, we replicate the experimental pipeline using SimSiam and compare the two related work in self-supervised representation learning.

II. Related Work

A. Self-Supervised Representation Learning

Self-supervised learning for visual representations has progressed rapidly through three broadly distinct paradigms. Contrastive methods, exemplified by SimCLR [1] and built on the InfoNCE objective of van den Oord et al. [2], learn by attracting differently augmented views of the same image while repelling views of all other images in the batch. SimCLRv2 [3], extended this framework with deeper projection heads and demonstrated that large self-supervised models become particularly effective when fine-tuned with limited labels. Khosla et al. [4] showed that the contrastive objective can be extended to incorporate label information when available which blurred the distinction between supervised and self supervised learning.

B. The STL10 Benchmark

The STL10 dataset [5] was introduced specifically as a benchmark for unsupervised and self-supervised feature learning. Its design, 5,000 labeled training images across 10 classes alongside 100,000 unlabeled images drawn from a wider distribution — directly mirrors the setting in which self-supervised learning is most useful: abundant unlabeled data and scarce labels. The 96×96 resolution offers more spatial detail than CIFAR-10 while remaining tractable for academic compute budgets. We follow common practice in downscaling to 32×32 to further reduce computational cost, while preserving the labeled/unlabeled split that motivates the benchmark.

C. Linear Evaluation Protocol

Evaluating learned representations by training a single linear classifier on top of a frozen encoder has become the standard in self-supervised learning. Kolesnikov et al. [6] systematically examined this protocol and demonstrated that linear probing reliably reflects representation quality across architectures and pretext tasks, while also exposing how strongly performance depends on factors such as encoder capacity and pretraining duration. Because the encoder is frozen and the classifier has minimal parameters, linear evaluation isolates the quality of the learned features from the capacity of the downstream model which is a property we exploit in Part 2 of our experiments by sweeping over label budgets from 10 to 500 images per class.

D. Data Augmentation in Self-Supervised Learning

The augmentation pipeline is widely recognized as the single most consequential design choice in contrastive learning. Chen et al. [1] showed that the composition of augmentations, particularly the combination of random cropping with color distortion drives the majority of SimCLR's performance, and that no single augmentation in isolation suffices. The intuition is that the encoder is forced to become invariant to whatever transformations its views differ by choosing augmentations therefore amounts to choosing which factors of variation the representation should ignore. This insight extends to negative-free methods such as BYOL [7] and SimSiam [8], which remain similarly sensitive to the augmentation distribution despite their structural differences from contrastive approaches. We adopt the standard SimCLR augmentation pipeline; crop-and-resize, horizontal flip, color jitter, grayscale, and Gaussian blur for both SSL methods to ensure that comparisons isolate the effect of the learning objective rather than the augmentation policy.

III. System Architecture

This project explores SSL through two contrastive frameworks – SimCLR and SimSiam — and evaluates the quality of their learned representations with a linear probe: logistic regression. Both methods train a convolutional encoder entirely without labels, relying instead on data augmentation to learn similar qualities. After pretraining, the encoder is frozen and a single linear classifier is fit on small

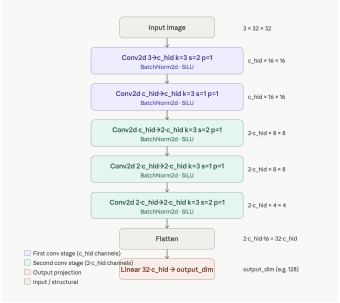


Fig. 1: Architecture diagram of the supervised baseline network.

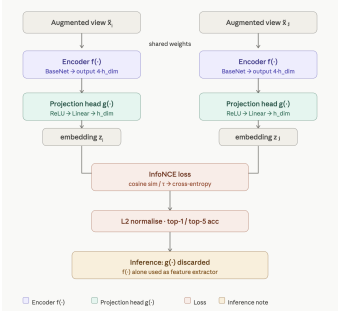


Fig. 2: Architecture diagram of the SimCLR model.

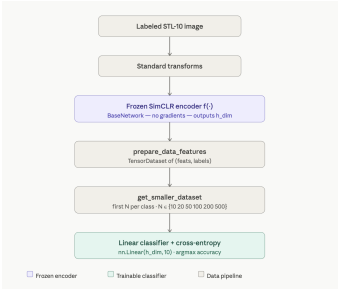


Fig. 3: Architecture diagram of the SimCLR + logistic regression pipeline.

labeled subsets, testing whether the unsupervised representations are genuinely useful for downstream recognition.

A. Supervised Baseline

The convolutional backbone is shared across all experiments. In the supervised baseline, it is trained end-to-end with cross-entropy loss on labeled STL-10 data, while SimCLR and SimSiam use the same architecture as their encoder f(·) but train it without any labels. Inside this backbone are five convolutional blocks with BatchNorm and SiLU activations that progressively halve the spatial resolution from 32x32 to 4x4 while doubling channel width, followed by a flatten and a linear projection to output_dim, as can be seen in Figure 1.

B. SimCLR

SimCLR [3] learns visual representations by contrasting two augmented views of the same image against views of all other images in the batch, as shown in Figure 2. Each view is passed through a shared base encoder $f(\cdot)$ which is a five-layer CNN with SiLU activations and batch normalization that downsamples 32×32 inputs to a 512-dimensional representation $\mathbf{h}$ and a two-layer MLP projection head $g(\cdot)$ with a $4\times$ wider hidden layer that maps $\mathbf{h}$ to a 128-dimensional embedding $\mathbf{z}$. The InfoNCE loss [2] with cosine similarity and temperature $\tau = 0.07$ pulls positive pairs together while pushing all $2N-2$ negatives apart. This is implemented by computing the full $2N \times 2N$ similarity matrix in a single operation and locating positive pairs via a rolled identity mask. After pretraining, the projection head is discarded and only $f(\cdot)$ is retained since representations from $\mathbf{h}$ transfer better than those from $\mathbf{z}$ [3].

C. SimCLR Logistic Regression

As seen in Figure 3, the frozen encoder extracts a feature vector for every labeled image via prepare_data_features, producing a TensorDataset that never touches raw pixels again. get_smaller_dataset then subsamples the first N examples per class, and a single nn.Linear layer is trained with cross-entropy on these features — testing how much useful structure SimCLR encoded without supervision.

D. SimSiam

Similar to SimCLR, SimSiam processes two augmented views through a shared encoder and projection head to produce z_1 and z_2, which can be seen in Figure 4. The key difference is an additional predictor MLP h(·) applied to each branch, whose output p is compared to the stop-gradient of the other branch's z. This asymmetry — one branch predicts, the other is treated as a fixed target — prevents model collapse without requiring negative pairs or a momentum encoder. Instead, it relies on the symmetrized cosine similarity loss -0.5 · (sim_1 + sim_2).mean().

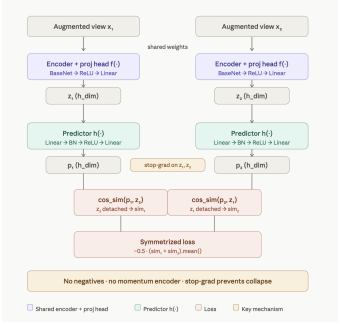


Fig. 4: Architecture diagram of the Sim Siam network.

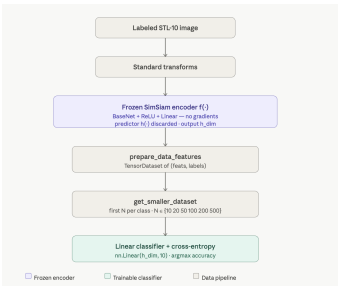


Fig. 5: Architecture diagram of the Sim Siam + logistic regression pipeline.

E. SimSiam Logistic Regression

This architecture is structurally identical to the SimCLR + logistic regression pipeline, as can be seen in Figure 5. The encoder and projection head are frozen, the predictor h(·) is discarded, and the same prepare_data_features and get_smaller_dataset utilities feed a linear classifier trained with cross-entropy. Comparing this curve against SimCLR's across the six different numbers of images per class reveals which method learned more transferable representations.

IV. Results and Validation

A. SimCLR

We evaluated all models using top-1 and top-5 validation accuracy. After just 10 epochs of training, validation accuracy increases approximately linearly, settling in the 50% to 70% range.

After 100 epochs validation accuracy begins to plateau, indicating the model is approaching saturation. Performance has improved to the 70% to 80% range, suggesting the model benefits substantially from additional training.

At 500 epochs, validation accuracy exhibits increased variability while remaining within the 80%

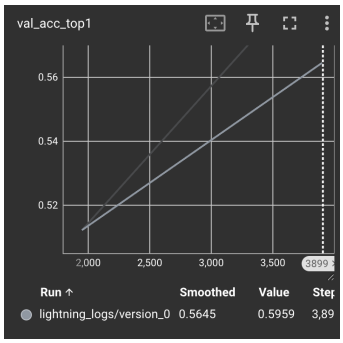


Fig. 6: SimCLR validation accuracy (top-1) for 10 epochs.

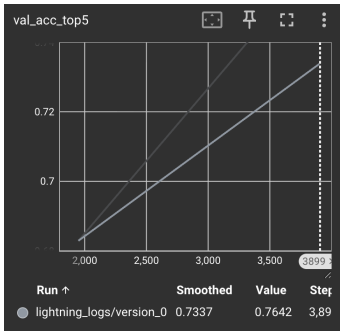


Fig. 7: SimCLR validation accuracy (top-5) for 10 epochs.

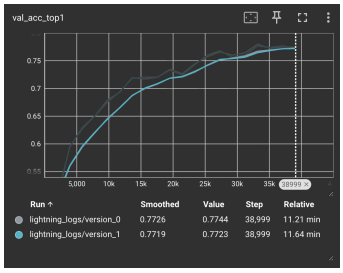


Fig. 8: SimCLR validation accuracy (top-1) for 100 epochs.

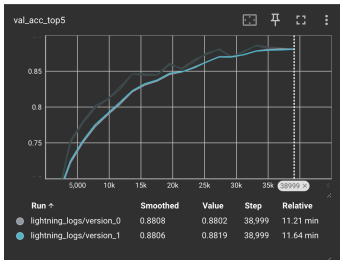


Fig. 9: SimCLR validation accuracy (top-5) for 100 epochs.

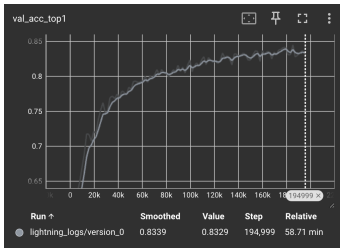


Fig. 10: SimCLR validation accuracy (top-1) for 500 epochs.

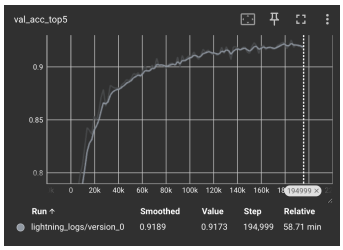


Fig. 11: SimCLR validation accuracy (top-5) for 500 epochs.

to 90% range. Validation loss decreases steadily throughout training and, despite some fluctuation, does not exhibit clear signs of overfitting.

B. SimCLR Logistic Regression

To assess whether the SimCLR encoder learned meaningful representations rather than superficial features, we evaluated its embeddings using logistic regression as a linear probe. Across all epoch configurations, performance generally improves with more labeled images per class, with the most pronounced gain occurring between 20 and 50 images per class — accuracy increases from approximately 40% to 50% across all three epoch trials. Performance continues to improve up to 500 images per class, though with diminishing returns.

At 10 epochs, the model reaches a test accuracy of approximately 57.5%, consistent with the

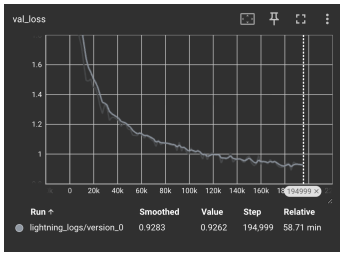


Fig. 12: SimCLR validation loss for 500 epochs.

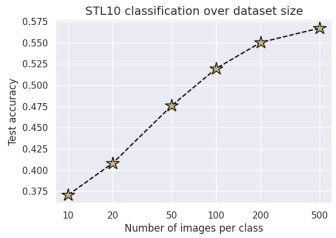


Fig. 13: SimCLR logistic regression linear probe for 10 epochs. The curve plateaus between 200 and 500 images per class, suggesting the encoded reaches its representational capacity for this configuration at approximately 200 images per class.



Fig. 14: SimCLR logistic regression linear probe for 100 epochs. The more linear learning curve suggests the model has not yet saturated at 500 images per class.

SimCLR validation accuracy reported above. This agreement between the two evaluations supports the conclusion that the model is learning generalizable representations rather than memorizing training examples.

With 100 epochs of pretraining, test accuracy reaches approximately 65% to 66%, which aligns with the lower end of the SimCLR validation results. As with the 10-epoch case, performance is maximized at 500 images per class, though the learning curve appears more linear than in the shorter training run, suggesting the encoder has not yet saturated at this data scale.

After 500 epochs of pretraining, test accuracy remains just above 65%, which is approximately 20% to 30% below the accuracy reported in the original SimCLR paper. This discrepancy likely reflects some degree of overfitting to the training distribution or a mismatch between the learned representations and the label structure used during linear evaluation.

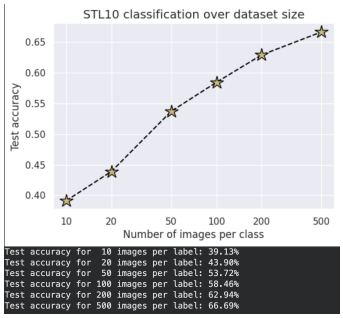


Fig. 15: SimCLR logistic regression linear probe for 500 epochs. The learning curve resembles the 10-epoch profile, and the encoder appears to begin saturating at 500 images per class.

C. Supervised Baseline

At 10 epochs, the supervised baseline achieves a test accuracy of 58.84%, while SimCLR with logistic regression achieves 56.19%. Although the baseline performs marginally better, the gap is small enough that the two approaches can be considered roughly equivalent at this training duration — a promising result that suggests SimCLR is learning comparably useful representations without any label supervision during pretraining.

At 100 epochs, the baseline reaches 67% test accuracy compared to 66% for SimCLR with logistic regression, a gap of just one percentage point. However, the baseline's training accuracy in the low 80%s indicates substantial overfitting, meaning the baseline's advantage over SimCLR is largely an artifact of memorization rather than superior generalization.

At 500 epochs, the baseline achieves 67.66% compared to 66.69% for SimCLR with logistic regression. While SimCLR does not surpass the supervised baseline, it continues to match it competitively. Notably, the baseline's training accuracy of 98.14% at this stage reveals severe overfitting, further contextualizing the similarity of test-set results and reinforcing the practical value of the contrastive learning approach.

D. SimSiam

SimSiam is a self-supervised learning framework that shares a backbone encoder architecture with SimCLR but introduces a stop-gradient operation on the target branch to prevent representational collapse. This design eliminates the need for large batch sizes, negative pairs, or momentum encoders, making SimSiam substantially more memory-efficient and simpler to implement than SimCLR.

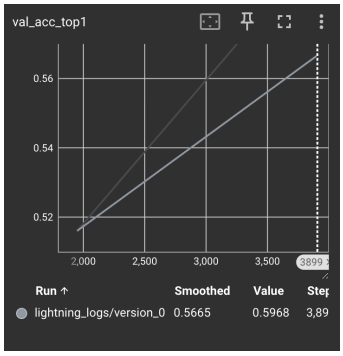


Fig. 16: SimSiam validation accuracy (top-1) for 10 epochs.

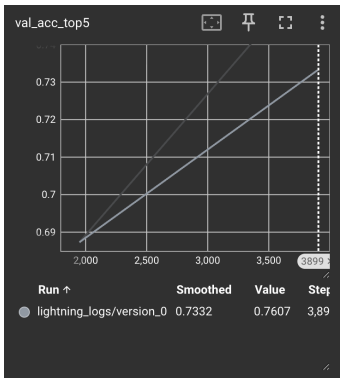


Fig. 17: SimSiam validation accuracy (top-5) for 10 epochs.

At 10 epochs, SimSiam produces results closely comparable to those of SimCLR, with validation accuracy following a similar trajectory throughout training.

By 100 epochs, SimSiam marginally outperforms SimCLR, reaching the 75% to 85% accuracy range. This improvement, achieved with lower computational overhead, highlights the efficiency advantages

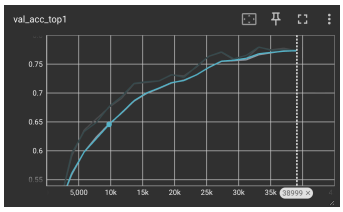


Fig. 18: SimSiam validation accuracy (top-1) for 100 epochs.

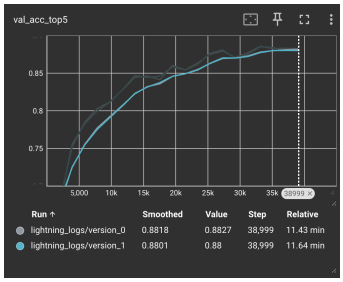


Fig. 19: SimSiam validation accuracy (top-5) for 100 epochs.

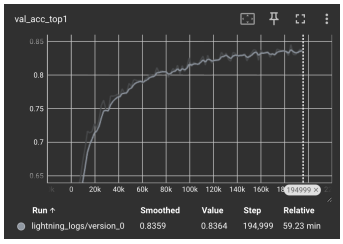


Fig. 20: SimSiam validation accuracy (top-1) for 500 epochs.

of the SimSiam architecture.

At 500 epochs, SimSiam continues to mirror SimCLR's behavior: validation accuracy plateaus with mild fluctuation, and validation loss declines steadily without clear signs of overfitting. These results confirm that SimSiam converges reliably over extended training.

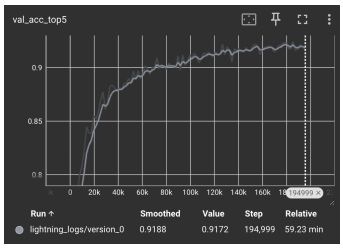


Fig. 21: SimSiam validation accuracy (top-5) for 500 epochs.

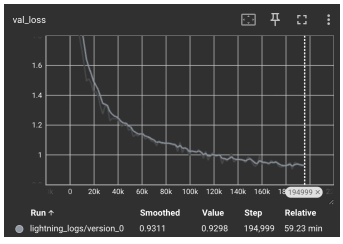


Fig. 22: SimSiam validation loss for 500 epochs.

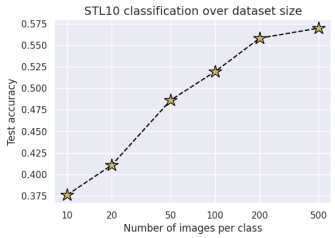


Fig. 23: SimSiam logistic regression linear probe for 10 epochs.



Fig. 24: SimSiam logistic regression linear probe for 100 epochs.

E. SimSiam Logistic Regression

The SimSiam linear probe results closely mirror those of SimCLR across all epoch configurations. The sharpest accuracy gains again occur between 20 and 50 images per class. As with SimCLR, the encoder saturates at approximately 500 images per class after 10 epochs of pretraining, transitions to a more linear scaling regime at 100 epochs, and begins to plateau again at 500 images per class after 500 epochs. Overall test accuracies align closely with the SimCLR logistic regression results, demonstrating that SimSiam achieves comparable representational quality at reduced computational cost. This is a meaningful result that validates the

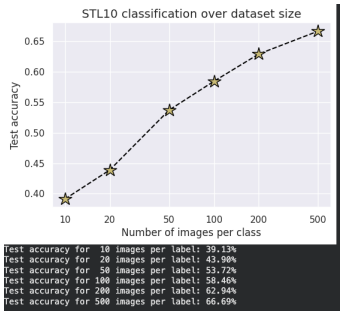


Fig. 25: SimSiam logistic regression linear probe for 500 epochs.

efficiency-accuracy tradeoff of the SimSiam design.

V. Individual Contributions

MB worked on the logistic regression models for both CL pipelines, supervised baseline model, and the SimSiam model. MB also wrote the results, architecture, and conclusion sections.

TB developed the SimCLR model, wrote the abstract, introduction, related work, and SimCLR architecture.

VI. Conclusion and Future Work

The core finding of this paper is that SimSiam matched SimCLR despite having a simpler and more memory-efficient architecture. Furthermore, both SSL methods competitively matched the fully supervised baseline, which was severely overfitting with a 98% training accuracy while the SSL models were not.

The SimCLR model in the original paper [1] received a 76.5% top-1 accuracy, whereas our model reached about 66% at 500 epochs on small labeled subsets, as can be seen in Figure 15. We argue that, given the smaller scale and complexity of our setup, these results are reasonably consistent. Our compact CNN achieved competitive SSL pretraining dynamics relative to the original SimCLR framework, as evidenced by the validation accuracy curves in Figure 10. We explain the gap between our logistic regression accuracy and the original paper's reported numbers as expected given our smaller encoder and more restricted evaluation, since our evaluation protocol intentionally restricted label budgets to probe label efficiency rather than maximize absolute accuracy.

Another takeaway is the jump between 20-50 images per class in the logistic regression. This suggests that the SSL representations encode genuinely useful structure even with very few labels, but that there is a threshold below which the linear classifier simply does not have enough signal to exploit those representations well. Above about 50 images per class, the classifier can start reliably reading the feature space the encoder learned. This suggests that the quality of the learned representations, rather than the capacity of the linear classifier, is the primary driver of downstream performance.

In future work, we would like to evaluate whether our SimCLR performs competitively when trained on the full dataset with greater compute resources. If it does, this would suggest that a compact CNN encoder is sufficient for competitive SSL performance. Additionally we will compare BYOL or Barlow Twins to our results for a more complete picture.

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